Towards Subseasonal-to-Seasonal Drought-to-Flood Prediction

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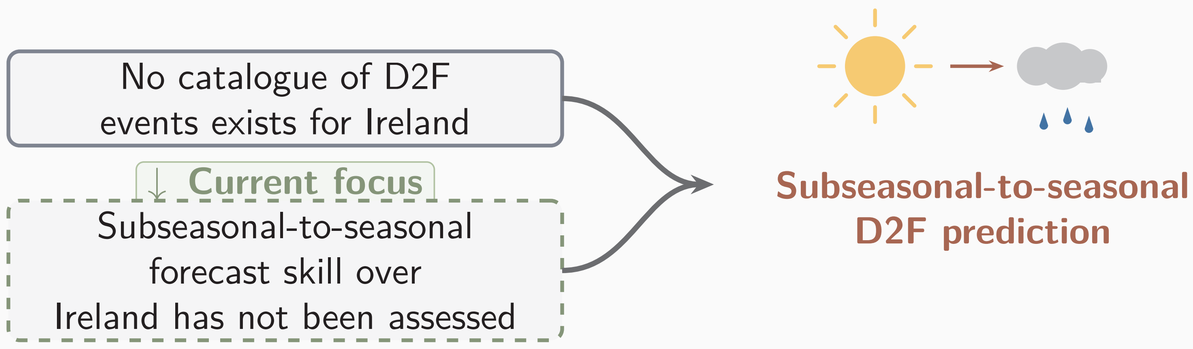
Introduction

Floods and droughts can occur in sequence — a **drought-to-flood transition (D2F)**. Globally, **~25% of floods are preceded by drought** [2], and socio-economic losses can be up to **8x greater** than from isolated hazards [3]. Forecasting D2Fs at subseasonal-to-seasonal lead times could support early action, but 2 research gaps remain ⟶



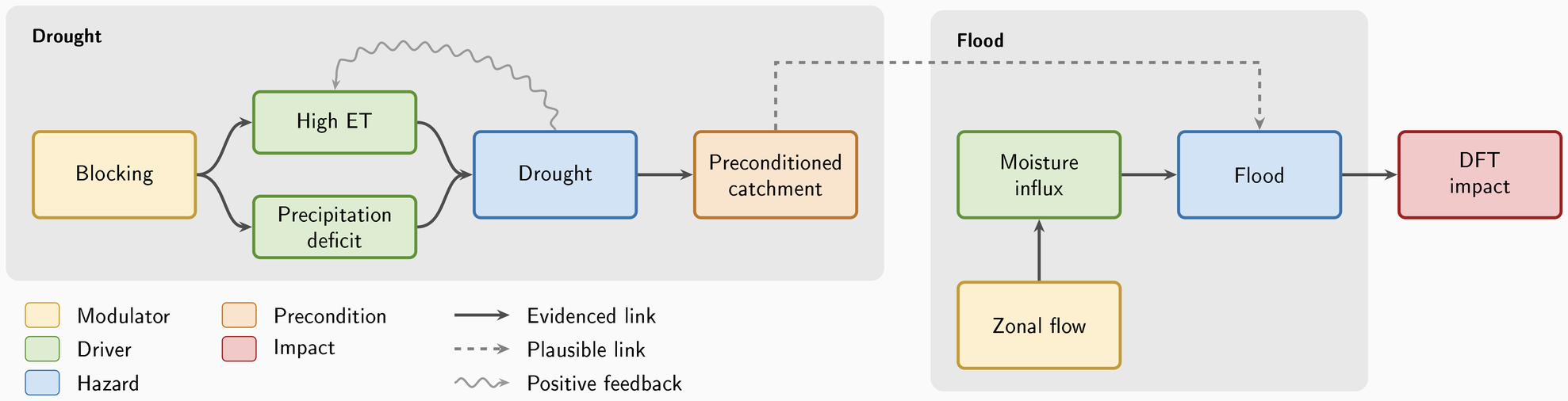


Figure 1: Physical pathway for D2F events, after the framework of Bevacqua et al. [1].

Methods

We verify 2 m temperature forecasts at **2–6 week** lead times, pooling all reforecast initialisations (2005–2024) per season and lead. Two complementary metrics are used: **Fair CRPSS** (Continuous Ranked Probability Skill Score) [4], scored against ERA5 climatology, and **centred ACC** (Anomaly Correlation Coefficient). Mean bias is removed by leave-one-year-out correction.

$$\mathrm{ACC} = \frac{\sum_{m=1}^{M} w_m (f'_m - \bar{f}')(o'_m - \bar{o}')}{\left[\sum_{m=1}^{M} w_m (f'_m - \bar{f}')^2 \cdot \sum_{m=1}^{M} w_m (o'_m - \bar{o}')^2\right]^{1/2}}$$

f'_m, o'_m = forecast/observed anomalies
$\bar{f}'$, $\bar{o}'$ = area-weighted spatial means
$w_m = \cos\varphi$ (latitude weight)
M = number of gridpoints

Wilks (2011), Eq. 7.59 [5]

Data

System	ECMWF ENS extended-range
IFS cycle	CY49R1 (2024)
Init. freq.	Every 2 days
Members	11 (1 control + 10 perturbed)
Horizon	Weeks 2–6
Variable	2 m temperature
Grid	0.75° regular lat-lon (MARS)
Reference	ERA5 + Met Éireann stations
Reforecasts	2005–2024

Results

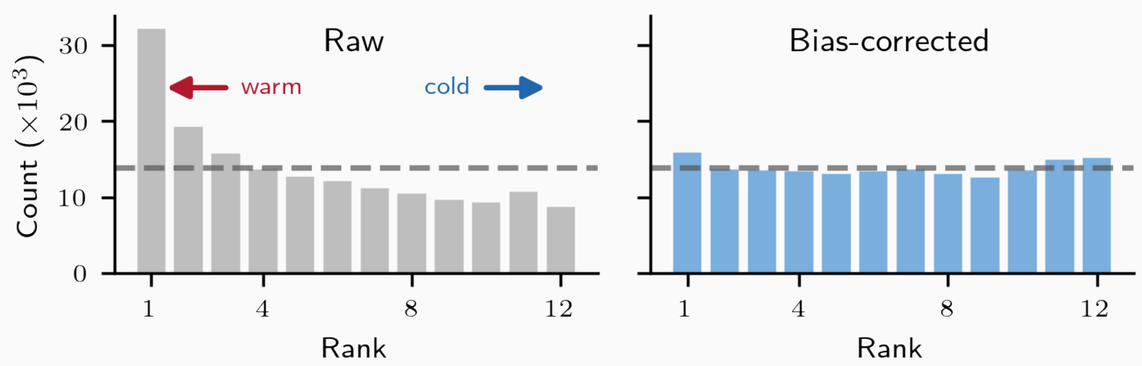


Figure 2: Rank histograms before/after seasonal LOYO bias correction.

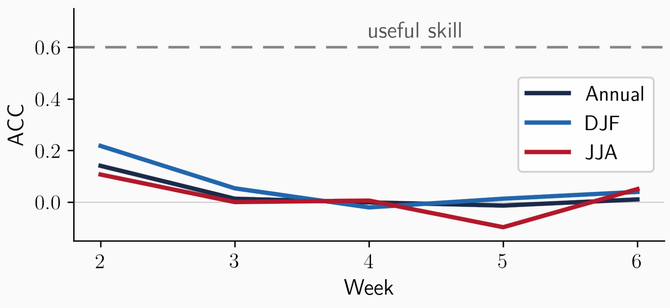


Figure 3: ACC vs lead time.

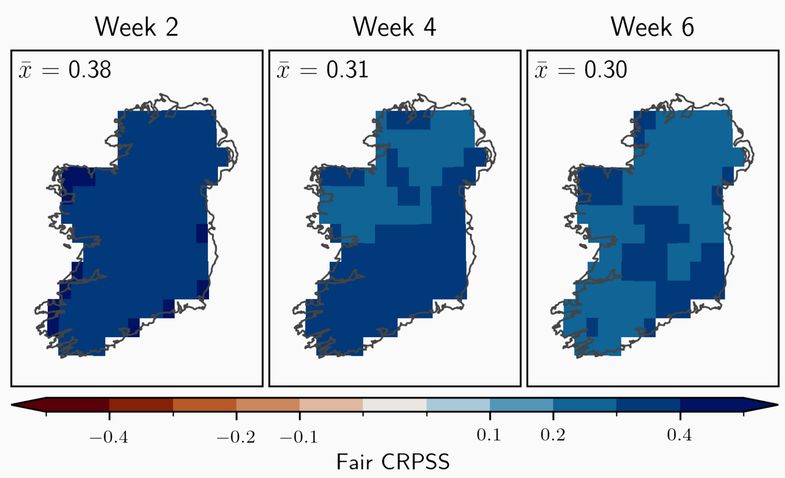


Figure 4: Fair CRPSS across lead time (bias-corrected).

Future Work

- Compare with SEAS5 seasonal forecasts to map skill across the full subseasonal-to-seasonal timescale
- Create D2F event catalogue for Ireland; link to catchment and soil properties

⟶ **Identify predictable at-risk catchments**

References

[1] Bevacqua et al. (2021). *Earth's Future*, 9, e2021EF002340. [2] Matanó et al. (2024). *Environ. Res. Lett.*, 19, 064048. [3] Worou & Messori (2025). *Environ. Res. Lett.*, 20, 104024. [4] Ferro (2014). *Q. J. R. Meteorol. Soc.*, 140, 1917–1923. [5] Wilks (2011). *Statistical Methods in the Atmospheric Sciences*, 3rd ed., Eq. 7.59.